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12 SEPTEMBER 1966
SUPERSEDING
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MILITARY SPECIFICATION

REGULATOR, PRESSURE, FLAMETHROWER

1. SCOPE

1.1 This specification covers one type of pressure regulator.

2. APPLICABLE DOCUMENTS

2.1 **Government documents.** The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-8585 — Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-130 — Identification Marking of U.S. Military Property.

MIL-STD-171 — Finishing of Metal and Wood Surfaces.

MIL-STD-810 — Environmental Test Methods for Aerospace and Ground Equipment.

AND 10050 — Bosses, Standard Dimensions for Gasket Seal Straight Thread.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification

(Application for copies of these ratings, rules, and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Ill. 60606.)

3. REQUIREMENTS

3.1 **Materials and components.**

3.1.1 *Materials.* All materials used in the manufacture of the regulator shall conform to the specifications forming a part of this specification. Materials not definitely specified shall be of an accepted commercial type and grade suitable for the purpose intended. Material shall be free from defects and imperfections that might affect serviceability of the finished product.

3.1.2 *Components.* Components shall be of an accepted commercial type or grade and of the manufacturer's latest design.

3.2 **Design.** The regulator shall be complete and capable of operation under the conditions specified in this specification. All

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regulators procured under this specification from one supplier shall be physically and mechanically identical (see figure 1). Regulators shall be of a design not hazardous to personnel. The regulator shall be capable of maintaining its rated functioning for 50,000 cycles (a cycle is defined as a 3 second flow of regulated air). The regulator shall have an adjustment feature to obtain the setting specified in 3.3. The adjustment shall be suitably designed to discourage tampering. The regulator shall be capable of operating at temperatures of −40°F to +155°F. The weight of the regulator including the relief valve shall not exceed 1.2 pounds.

3.3 **Functioning.** The regulator shall maintain an outlet pressure of 285 ± 45 pounds per square inch gage (psig) at a flow rate of 0 to 40 standard cubic feet per minute (scfm) (60°F at 14.7 psi) of clean air or nitrogen at inlet pressures of 675 (+0 or −50) psig and 2,000 (+50 or −0) psig when tested as specified in 4.3.4.1.

3.4 **Relief valve.** The regulator relief valve shall not relieve at a pressure less than 345 psig and shall relieve with a minimum flow rate of 10 scfm at a maximum pressure of 400 psig when tested as specified in 4.3.4.2.

3.5 **Leakage.** The regulator shall not leak more than 15 cubic centimeters per minute of air or nitrogen when tested as specified in 4.3.4.3.

3.6 **Preproduction.** Prior to the start of regular production, a preproduction sample of regulators shall be produced in accordance with this specification for examination and tests (see 4.2 and 6.2).

3.6.1 *High temperature.* The regulator shall conform to 3.3, 3.4 and 3.5 when tested as specified in 4.2.3.1.

3.6.2 *Low temperature.* The regulator shall conform to 3.3, 3.4 and 3.5 when tested as specified in 4.2.3.2.

3.6.3 *Temperature shock.* The regulator shall conform to 3.3, 3.4 and 3.5 when tested as specified in 4.2.3.3.

3.6.4 *Salt fog.* The regulator shall conform to 3.3, 3.4 and 3.5 when tested as specified in 4.2.3.4.

3.6.5 *Sand and dust.* The regulator shall conform to 3.3, 3.4 and 3.5 when tested as specified in 4.2.3.5.

3.6.6 *Proof pressure.* The regulator body shall not leak when subjected to a minimum pressure of 3,000 psig as specified in 4.2.3.6 (see 6.3).

3.6.7 *Burst pressure.* The regulator body shall not rupture or burst when subjected to a minimum pressure of 5,000 psig as specified in 4.2.3.7 (see 6.3).

3.6.8 *Cyclic functioning.* The regulator shall conform to 3.3 when cycled in accordance with table I and tested as specified in 4.2.3.8.

TABLE I. *Criteria for failure rate testing*

Column I Number of failures	Column II Cumulative time (cycles) (see 3.2)
1	30,000
2	48,000
3	63,000
4	78,000
5	92,000
6	105,000
7	119,000
8	132,000
9	145,000
10	157,000

The above cumulative operating time in column II shall be met before the indicated number of failures in column I has occurred. The regulator shall demonstrate a mean life of at least 10,000 cycles, at a 95 percent confidence level.

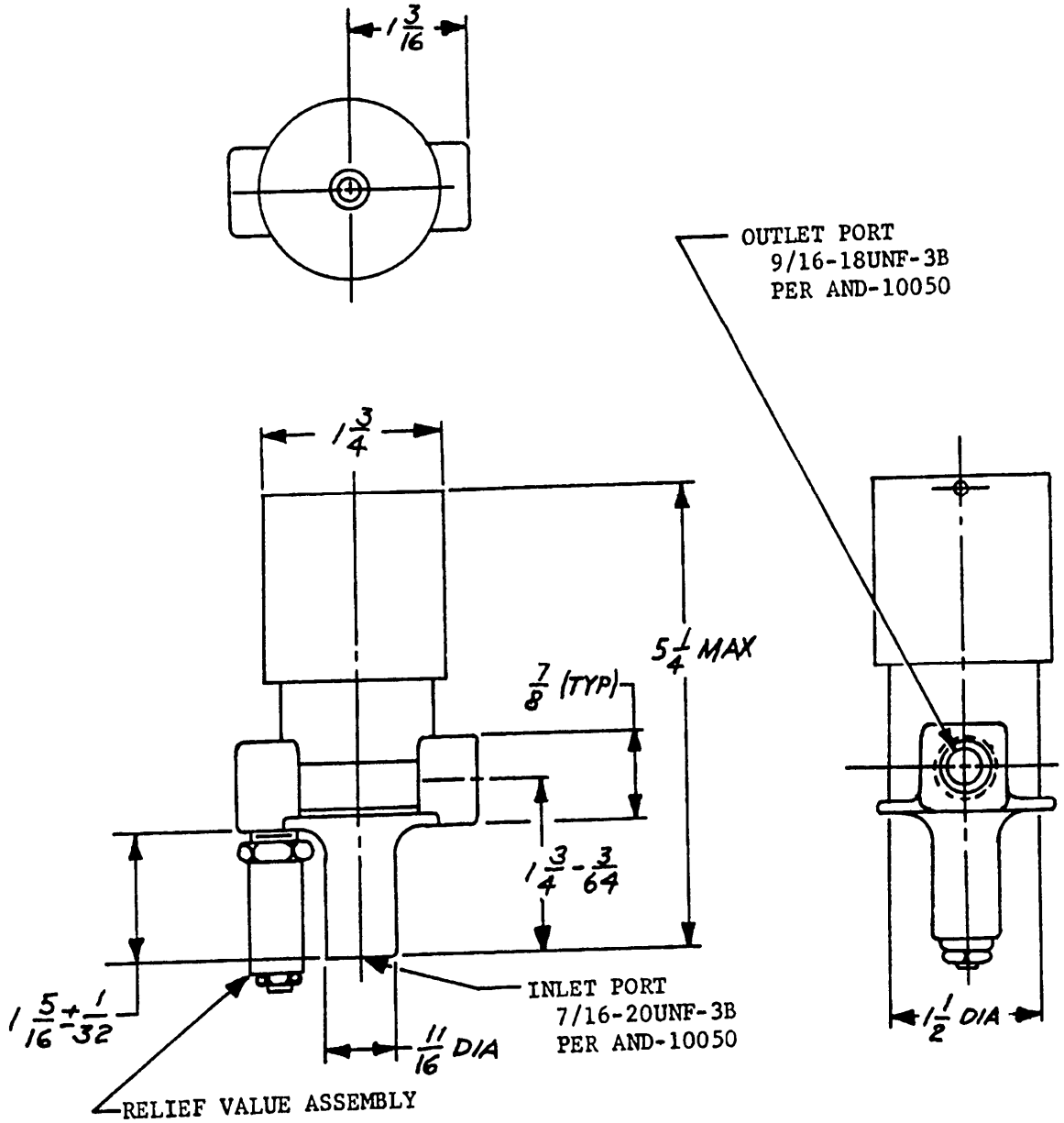
3.6.9 *Vibration.* The regulator shall conform to 3.3, 3.4, and 3.5 after testing as specified in 4.2.3.9.

3.6.10 *Mechanical shock.* The regulator shall conform to 3.3, 3.4, and 3.5 after testing as specified in 4.2.3.10.

3.7 **Identification.** The regulator shall be identified by a nameplate in accordance with MIL–STD–130.

3.8 **Workmanship.** The regulator shall be free from foreign matter (oil or grease) and damage such as cracks or deformation which could affect its use.

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NOTES:

1. DIMENSIONS ARE IN INCHES.
2. PROTECTIVE FINISH FOR EXTERIOR SURFACES: SURFACE TREATMENT NO.7.1.2 OR 7.2.2 COLOR OLIVE DRAB, FOLLOWED BY SEMI-GLOSS PAINT SYSTEM *21.19 COLOR NO. X24087, MIL-STD-171 FOR THE ARMY OR MARINE CORPS AS SPECIFIED.

* PRIMER SHALL BE COLOR T, PER SPEC MIL-P-8585

Figure 1.- Regulator assembly

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 *Supplier's responsibility.* Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 *Objective evidence.* The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Preproduction.

4.2.1 *Sample.* A sample of 63 regulators shall be produced using the same materials, methods, equipment, processes, and procedures as will be used in regular production. Any change in materials, methods, processes, or procedures will require a new preproduction sample.

4.2.2 *Inspection.*

4.2.2.1 *For examination and nondestructive tests.* The sample regulators shall be examined and tested in accordance with the classification of defects and MIL–STD–105 (4.3.3.3).

4.2.2.2 *For destructive tests.* The regulators (4.2.1) shall be sampled and tested in accordance with table II.

4.2.3 *Tests.*

4.2.3.1 *High temperature.* The regulator shall be tested in accordance with method 501.1 of MIL–STD–810.

4.2.3.2 *Low temperature.* The regulator shall be tested in accordance with method 502.1 of MIL–STD–810.

TABLE II. *Preproduction sampling and inspection acceptance/rejection criteria*

Quantity	Requirement	Test	Acceptance	Rejection
10	3.6.1	4.2.3.1	0	1
10	3.6.2	4.2.3.2	0	1
10	3.6.3	4.2.3.3	0	1
3	3.6.4	4.2.3.4	0	1
3	3.6.5	4.2.3.5	0	1
10	3.6.6	4.2.3.6	0	1
5	3.6.7	4.2.3.7	0	1
2(min)	3.6.8	4.2.3.8	*	*
5	3.6.9	4.2.3.9	0	1
5	3.6.10	4.2.3.10 (a) and (b)	0	1

* See table I.

4.2.3.3 *Temperature shock.* The regulator shall be tested in accordance with method 503.1 of MIL–STD–810.

4.2.3.4 *Salt fog.* The regulator shall be tested in accordance with method 509.1 of MIL–STD–810.

4.2.3.5 *Sand and dust.* The regulator shall be tested in accordance with method 510.1 of MIL–STD–810.

4.2.3.6 *Proof pressure.* The relief valve shall be removed from the regulator body and the outlet and relief ports plugged. The inlet shall be connected to a regulated hydrostatic pressure. The pressure shall be adjusted in accordance with 3.6.6 and maintained for a minimum period of 1 minute. The body shall be observed for leaks.

4.2.3.7 *Burst pressure.* The relief valve shall be removed from the regulator body and the outlet and relief ports plugged. The inlet shall be connected to a regulated hydrostatic pressure. The pressure shall be adjusted in accordance with 3.6.7 and maintained for a minimum period of 1 minute. The body shall be observed for evidence of rupture or bursting.

4.2.3.8 *Cyclic functioning.* The regulator shall be mounted in a suitable fixture and cycled a minimum of 10,000 times in accordance with 3.6.8.

4.2.3.9 *Vibration.* The regulator shall be mounted in a rigid fixture which simulates

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the regulator installation. The fixture with the regulator shall be securely fastened to the vibration table and vibrated in accordance with figure 514–6, curve A and time table 514–II, schedule IV of MIL–STD–810, method 514.1.

4.2.3.10 *Mechanical shock.*

(a) *Regulator.* The regulator shall be mounted in a fixture which simulates the regulator installation. The regulator shall be subjected to three shocks in each direction applied along three mutually perpendicular axes for a total of 18 shocks. The shock pulse shall be in accordance with figure 516.1 of MIL–STD–810 with a peak value (A) of 15 g and nominal duration (D) of 11 milliseconds.

(b) *Repair parts.* The packaged regulator (see 6.4) shall be mounted on the test plate and each of the six sides shall be subjected to three shocks in each direction along the three mutually perpendicular axes for a total of 18 shocks. The shock pulse shape shall be as above (4.2.3.10(a)) with a peak value (A) of 30 g and a nominal duration (D) of 11 milliseconds.

4.2.3.11 *Weight.* The weight of the regulator shall be verified by a weighing device accurate to 0.10 pound.

4.3 Regular production.

4.3.1 *Lotting.* A lot shall consist of the regulators produced by one manufacturer, from the same materials, under essentially the same manufacturing conditions.

4.3.2 *Sampling.* Sampling for examination shall be conducted in accordance with MIL–STD–105.

4.3.3 *Inspection.*

4.3.3.1 *For examination.* Sample regulators shall be examined in accordance with the classification of defects and MIL–STD–105.

4.3.3.2 *Critical defects.* Each regulator shall be tested for the critical characteristics listed in the classification of defects.

4.3.3.3 *Classification of defects.*

(a) *Regulator, pressure, flamethrower.*

Categories	Defects	Acceptance standards
Critical:		
1	Functioning	4.3.4.1
2	Relief	4.3.4.2
3	Leakage	4.3.4.3
Major:	AQL 1.0 percent defective	
101	Envelope dimensions incorrect (figure 1)	
102	Threads incorrect	
103	Protective finish missing or incorrect	
104	Workmanship (3.8)	

4.3.4 *Tests.*

4.3.4.1 *Functioning.* The inlet of the regulator shall be connected to a regulated source of air and a suitable pressure gage. The outlet of the regulator shall be connected to a suitable pressure gage, a metering valve and an airflow meter. The inlet pressure shall be regulated to 675 with a tolerance of plus 0, minus 50 psig and the metering valve adjusted to flow rates of 0 and 40 scfm. The outlet pressures shall be recorded. After completion of the above tests, the inlet pressure shall be regulated to 2,000 with a tolerance of plus 50, minus 0 psig and the metering valve adjusted to flow rates of 0 and 40 scfm. The outlet pressure shall be recorded (see 6.3).

4.3.4.2 *Relief valve.* The outlet of the regulator shall be connected to a regulated source of air, a suitable pressure gage, and a calibrated rotometer. The inlet of the regulator shall be plugged. The air pressure shall be applied and slowly increased until the regulator starts to relieve. The pressure gage reading shall be recorded. The pressure shall be increased until the rotometer indicates 10 scfm. The pressure gage reading shall be recorded (see 6.3).

4.3.4.3 *Leakage.* The regulator outlet shall be plugged. The regulator shall be placed in a container capable of being sealed against

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leakage. The container shall have an inlet connected to a source of regulated air and a suitable pressure gage and an outlet connected to a calibrated rotometer. The regulator inlet port shall be connected to the container inlet. The container shall be sealed. The inlet pressure shall be regulated 2,000 plus 50, minus 0 psig and maintained for a minimum of 1 minute. The rotometer reading shall be recorded.

5. PREPARATION FOR DELIVERY (For repair parts packaging, see 6.4.)

5.1 Packaging, level C. The regulator shall be packaged to provide adequate protection against corrosion, deterioration, and damage from the supply source to the first receiving activity for immediate use.

5.2 Packing, level C. Regulators, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Uniform Freight Classification rules or the regulations or other common carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 Intended use. The regulator is intended for use with the M9–7 portable flamethrower and is used to regulate air from the high pressure sphere to the gun.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Preproduction:

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.

(2) Name and address of test facilities and shipping instructions when testing is performed by the Government.

(3) Time required for the Government to notify the supplier whether or not to proceed with production.

(c) Color required by user.

6.3 Caution. Due to the hazards inherent in testing with high pressure, suppliers are hereby notified that every precaution possible should be taken to prevent injury to personnel engaged in testing.

6.4 Repair parts packaging. When this item is being procured for stockage and issue as a repair part, the contract or order should specify that the item be preserved, packaged, and packed as shown on the repair parts packaging data sheet covering the item. The packaging data sheet should be ordered from the Technical Support Directorate, ATTN: SMUEA–TSE–DD, U.S. Army Edgewood Arsenal, Edgewood Arsenal, Md. 21010. The packaging data sheet should be identified by the Federal stock number (FSN) of this item.

6.5 International interest. Certain provisions of this specification are the subject of international standardization agreement. When amendment, revision, or cancellation of this specification is proposed, the departmental custodian will inform the Departmental Standardization Office so that appropriate action may be taken respecting the international agreement concerned.

Custodian:
Army—MU

International interest (see section 6)

Preparing activity:
Army—MU (EA)

Project No. 1040–A060

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